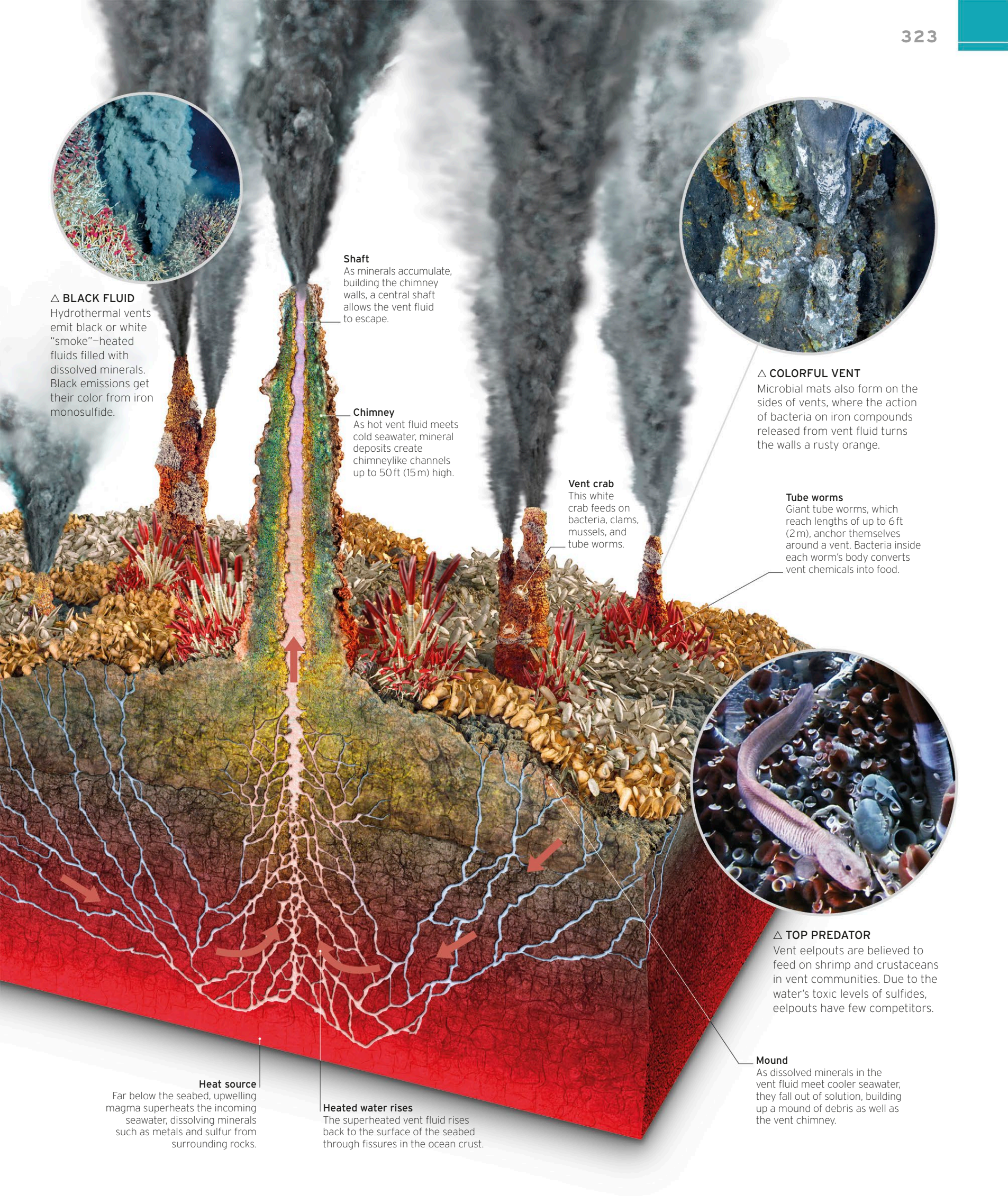


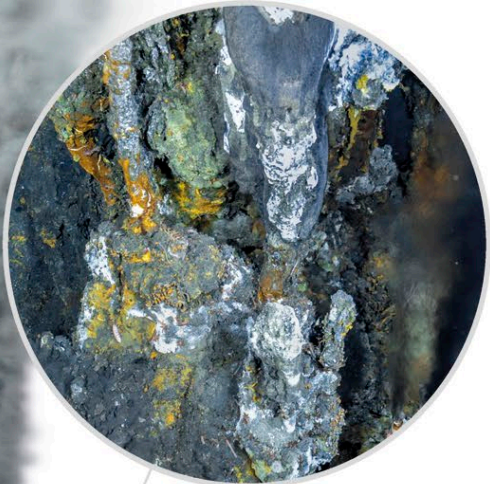


**△ BLACK FLUID**  
Hydrothermal vents emit black or white "smoke"—heated fluids filled with dissolved minerals. Black emissions get their color from iron monosulfide.

**Shaft**  
As minerals accumulate, building the chimney walls, a central shaft allows the vent fluid to escape.

**Chimney**  
As hot vent fluid meets cold seawater, mineral deposits create chimneylike channels up to 50 ft (15 m) high.

**Vent crab**  
This white crab feeds on bacteria, clams, mussels, and tube worms.



**△ COLORFUL VENT**  
Microbial mats also form on the sides of vents, where the action of bacteria on iron compounds released from vent fluid turns the walls a rusty orange.

**Tube worms**  
Giant tube worms, which reach lengths of up to 6 ft (2 m), anchor themselves around a vent. Bacteria inside each worm's body converts vent chemicals into food.



**△ TOP PREDATOR**  
Vent eelpouts are believed to feed on shrimp and crustaceans in vent communities. Due to the water's toxic levels of sulfides, eelpouts have few competitors.

**Heat source**  
Far below the seabed, upwelling magma superheats the incoming seawater, dissolving minerals such as metals and sulfur from surrounding rocks.

**Heated water rises**  
The superheated vent fluid rises back to the surface of the seabed through fissures in the ocean crust.

**Mound**  
As dissolved minerals in the vent fluid meet cooler seawater, they fall out of solution, building up a mound of debris as well as the vent chimney.